

WHEN SOFT MEANS MALIGNANT: LIMITATIONS OF ELASTOGRAPHY IN MUCINOUS BREAST CARCINOMAS

A COHORT STUDY WITH RADIOLOGIC-PATHOLOGIC CORRELATION

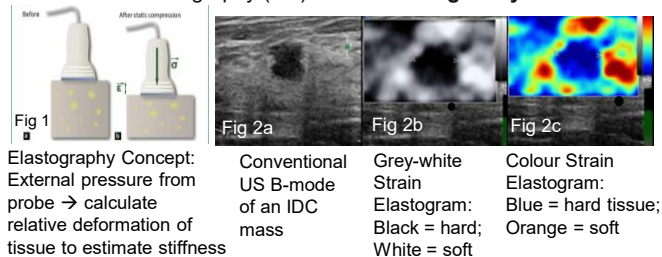
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BACKGROUND

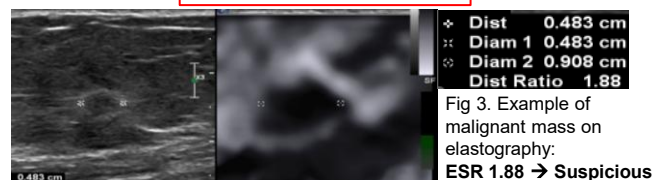
- Ultrasound elastography (UE) assumes **Malignancy = Hard Tissue**



- Also, cancers usually cause stiff surrounding desmoplastic tissue
→ **Elastographic size ratio (ESR)** becomes **enlarged** (fig 3)

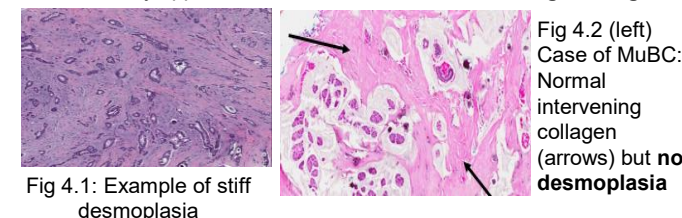
$$\text{Elastographic size ratio (ESR)} = \frac{\text{Length of lesion on elastogram}}{\text{Length of lesion on B mode US}}$$

ESR >1.1 = suspicious¹



Mucinous Breast Cancers (MuBC)

- Typically have minimal desmoplastic response (Fig 4.2)
- Abundant soft extracellular mucin²
- Hence may appear soft with normal ESR → **false neg elastogram**



OBJECTIVES

1. To evaluate the reliability of strain elastography in MuBC
2. To identify factors influencing accuracy of elastography in MuBC

METHODS

- Multi-institution retrospective study in Singapore
- 320 histologically confirmed MuBC cases from the SingHealth Joint Breast Cancer Registry and radiology records.
- Retrospective review of a **90** cases with available ultrasound strain elastography records from the year 2000 – 2020.
- **Subgroup analysis** – pure MuBCs vs mixed MuBCs
- Assessment categories:

Quantitative = **ESR >1.1 = suspicious; ≤1.1 = benign**

Qualitative = **Strain Colour Patterns: hard vs soft/intermediate**

RESULTS

Sensitivity of ESR
(>1.1 cutoff = Suspicious)

Sensitivity for All MuBCs

- 54/90 (60%) cases positive
- 36/90 (40.0%) were negative
- Sensitivity 60.0%

Sensitivity of Pure MuBCs vs Mixed MuBCs

- Pure MuBC 56.3% (45/80)
- Mixed MuBC 90.0% (9/10) (p = 0.046)

Sensitivity of Elastographic Strain Pattern (Hard Appearance = Suspicious)

Sensitivity for All MuBCs

- 74.4% (67/90)

Sensitivity of Pure MuBCs vs Mixed MuBCs

- 76.3% (61/80) Pure MuBCs with or w/o DCIS)
- 60.0% (6/10) in Mixed MuBCs (p=0.270)

DISCUSSION

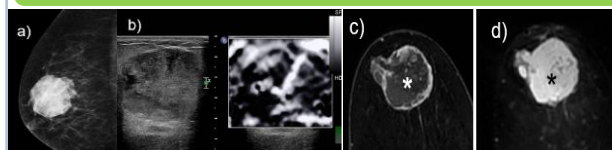
- Pathology review showed pure MuBCs did not elicit significant desmoplasia.
- Hence only 60% had enlarged ESR compared to >96% for other invasive breast cancers in other historical series.^{1,4,5,6}
- Enlarged ESR observed mainly in masses with histological spiculation - commonly in mixed MuBC.
→ Attributable to desmoplasia caused by non-mucinous invasive component.
- In MuBCs with DCIS, the DCIS component elicited hard sclerotic tissue which could explain the enlarged ESR in these cases⁷

CONCLUSION

- **Soft cancer masses without desmoplasia** are a **pitfall** of breast elastography, particularly in pure MuBCs
- UE findings should be interpreted in conjunction with conventional ultrasound and mammogram features
- Imaging features suggestive of MuBC include hyperdense mass on MG, isoechoic mass on US, ill-defined margins ± clustered cals
- **Suspicious B-mode / mammogram features should prompt biopsy regardless of elastographic softness**

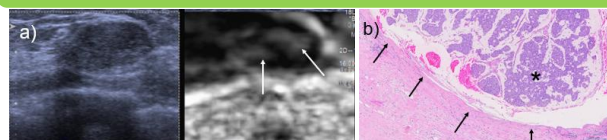
CASE ILLUSTRATIONS WITH MULTI-MODALITY CORRELATION

Case 1: Pure MuBC with abundant mucin content appearing soft on UE



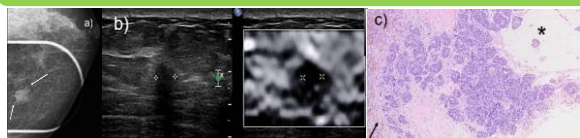
- a) Mammogram : large lobulated hyperdense mass.
- b) B-mode US (left pic) : isoechoic rounded mass.
UE (right pic) : heterogeneously white strain pattern
ESR = 1.04 (< 1.1).
- c) MRI post-contrast T1w with fat saturation shows mass is largely non-enhancing (*); only a thin rim of enhancement at periphery.
- d) MRI T2w STIR shows the non-enhancing component has hyperintense T2w signal due to the abundant water-rich mucin content (*). → explains why the mass appears soft.

Case 2: Pure MuBC with appearing predominantly soft on UE



- a) B-mode US (left pic): oval isoechoic mass with well-defined borders lacking spiculations.
UE (right pic): extensive soft, white areas (arrows). ESR < 1.1
- b) Haematoxylin and Eosin-stained section (50x magnification, 5x objective) : loose network of malignant cells (*) floating within extracellular mucin → soft appearance on UE.
The "pushing" borders (arrows) lack stromal reaction – explains the circumscribed margins on B-mode and non-enlarged ESR on UE.

Case 3: Mixed MuBC with IDC appearing hard on UE with enlarged ESR



- a) Spot compression mammogram: spiculated irregular mass (arrows)
- b) B-mode US (left pic) shows spiculated, tall, hypoechoic mass. Compared to B-mode, the size on UE (right pic) appears larger: ESR = 2.15. Also appears hard (predominantly black) on UE.
- c) Haematoxylin and Eosin stained section (50x magnification, 5x objective) shows mucinous carcinoma with small pool of mucin (*) and ductal carcinoma (left of mucin).
Infiltrative border with stromal desmoplastic reaction (arrow) contributes to increased stiffness → enlarged ESR.

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